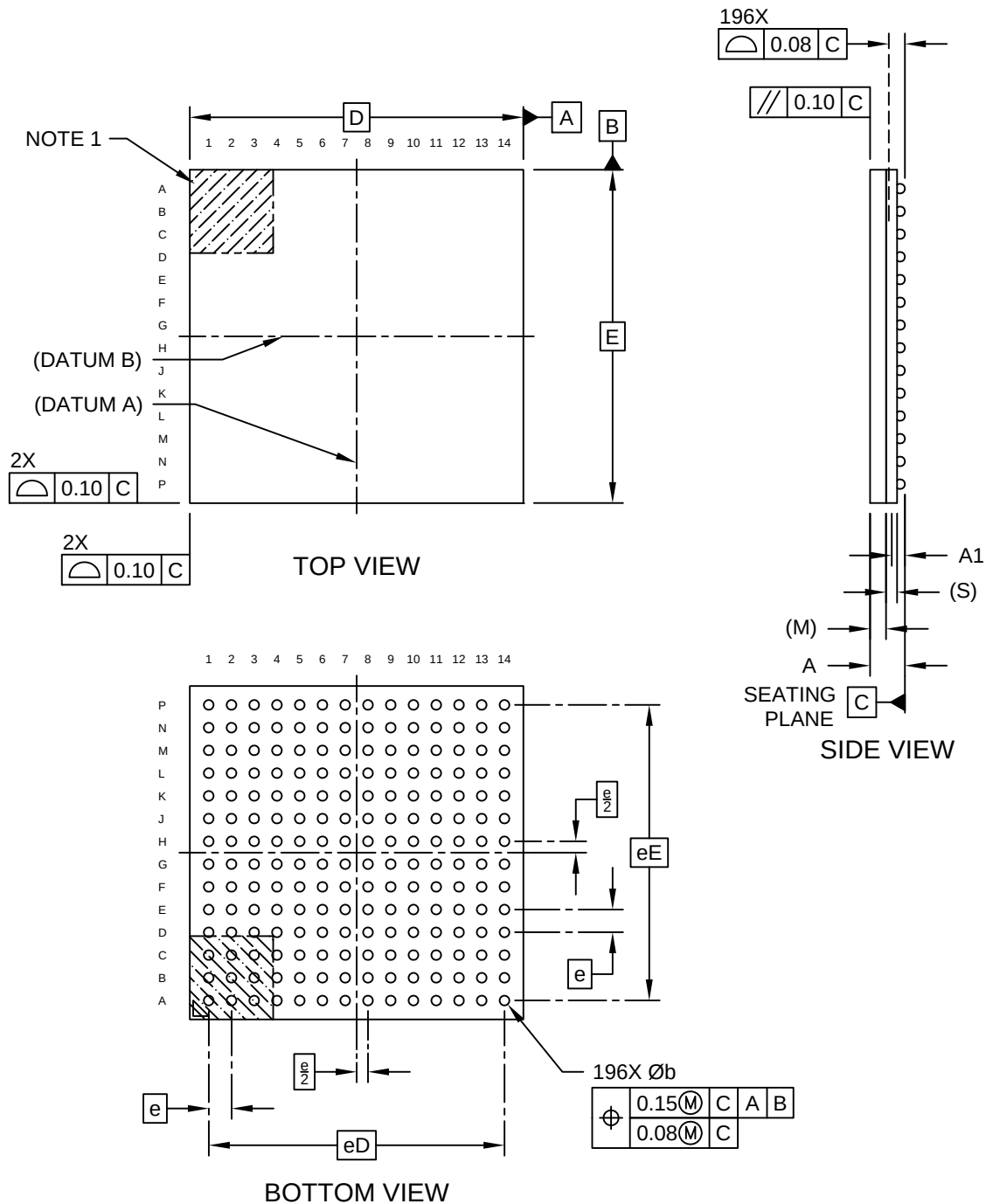


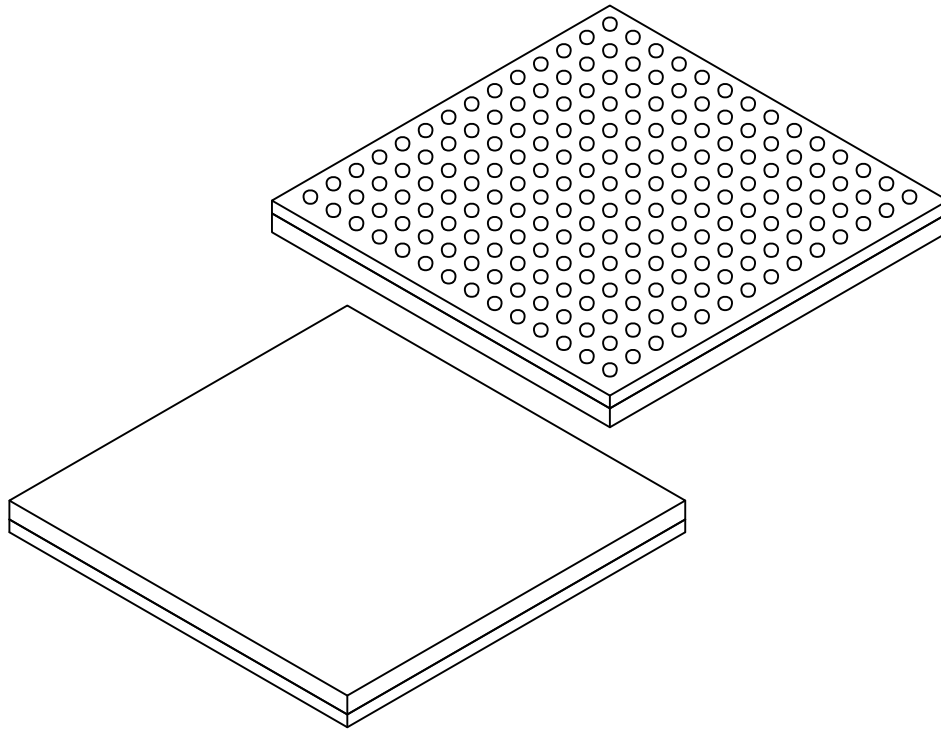
196-Ball Thin Fine Pitch Ball Grid Array (BAB) - 11x11 mm Body [TFBGA] Atmel Legacy Global Package Code GCX

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**196-Ball Thin Fine Pitch Ball Grid Array (BAB) - 11x11 mm Body [TFBGA]
Atmel Legacy Global Package Code GCX**

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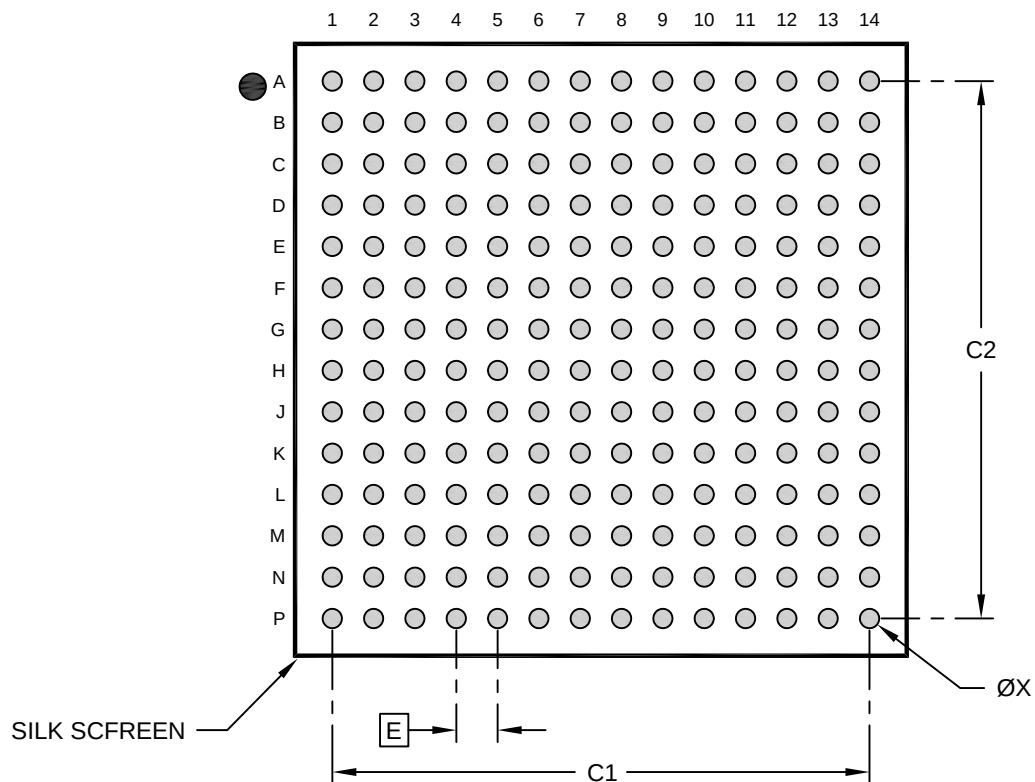
Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	196		
Pitch	e	0.75 BSC		
Overall Height	A	-	-	1.20
Ball Height	A1	0.16	0.25	0.26
Substrate Thickness	S	0.36 REF		
Mold Cap Thickness	M	0.53 REF		
Overall Length	D	11.00 BSC		
Overall Ball Pitch	eD	9.75 BSC		
Overall Width	E	11.00 BSC		
Overall Ball Pitch	eE	9.75 BSC		
Ball Diameter	b	0.27	0.32	0.37

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

196-Ball Thin Fine Pitch Ball Grid Array (BAB) - 11x11 mm Body [TFBGA] Atmel Legacy Global Package Code GCX

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.75 BSC		
Contact Pad Spacing	C1		9.75	
Contact Pad Spacing	C2		9.75	
Contact Pad Diameter (X196)	X			0.35

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23141 Rev A